Samit Mohan

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Experience

Beltech AI | *PyTorch, YOLO, DeepSORT, Triton, TensorRT, Docker* June 2025 – Present
Computer Vision Engineer Bangalore, Karnataka

- Designed and deployed an **end-to-end vision system** processing live RTSP streams across **10+ detection modules** (LPR, phone, helmet, wrong-way detection) achieving **0.9+ F1** across all modules for Indian CCTV infrastructure
- Cut inference latency by **54%** to **≤10 ms** on edge devices by replacing anchor-based detection with an NMS-free YOLOv10 architecture and optimizing Triton/TensorRT serving pipelines; sustained **0.90+ mAP** across 8 classes
- Built a **continuous learning pipeline** that ingests misclassifications from live traffic on a 2-week cadence, retrains, validates via **mAP50 quality gate**, and hot-swaps models on Triton with **zero downtime**
- Shipped vehicle-attribute microservices (color, type classification) at **0.92+ Macro-F1** with Grafana monitoring, structured logging, and A/B testing for model rollouts

MakeMyTrip | *Scala, Redis, MySQL, Kafka, AWS* August 2022 – October 2022
Machine Learning Intern Gurgaon, Haryana

- Built a **personalization API** using **multi-armed bandit algorithms** (epsilon-greedy, Thompson Sampling, UCB) to optimize homepage engagement; benchmarked strategies on custom datasets and shipped the best-performing variant to production
- Developed an NLP-based **resume screening system** using text similarity and keyword extraction for Data team hiring

Projects

MiniTorch | *Python, NumPy, CuPy*

- Built a **PyTorch-like deep learning framework from scratch** with reverse-mode autodiff, topological-sort computation graphs, and optional **GPU acceleration via CuPy**
- Implemented a full **Module system** – Linear, Conv2d, BatchNorm, Dropout, MaxPool2d, Adam/SGD optimizers, and LR schedulers (StepLR, CosineAnnealing)
- Trained custom CNN and MLP models achieving **95%+ MNIST accuracy**; verified gradient correctness with numerical finite-difference checks

Autoresearch-search | *Python, PyTorch, LLM Agents, FastAPI, SQLite*

- Built an **autonomous ML research framework** that uses LLM agents to iteratively generate, validate, and evaluate PyTorch training code in sandboxed subprocesses with **multi-seed evaluation** and **top-k search** with stagnation-aware exploration
- Designed a **pluggable task abstraction** where tasks expose a fixed API, baseline code, and evaluation function, enabling new ML benchmarks without modifying the core loop, controller, or database schema
- Implemented **live experiment monitoring** with a FastAPI/HTMX dashboard and Plotly HTML reports showing metric progression, failure analysis, experiment diffs, and category breakdowns backed by WAL-mode SQLite with per-experiment cost tracking

TensorTonic | *Python, NumPy, PyTorch, Markdown*

- Authored **200+ ML implementation problems** (gradient descent, optimizers, loss functions, attention) and research paper walkthroughs for an **ML interview prep platform**
- Built study plans and quizzes on ML/AI concepts for premium users; platform grew to **25,000+ signups** and **500+ paid users**

Teaching & Writing

InterviewReady – AI Teaching Assistant | *Python, PyTorch, LLMs* Dec 2025 – Present

- Designed and taught an AI engineering curriculum for **200+** software engineers in a live cohort with technical blog posts reaching **20,000+** readers; restructured course content for **10,000+** subscribers

Education

VTU, NHCE 2020 – 2024
B.Tech in Information Technology **(CGPA: 9.4/10.0)** Bangalore, Karnataka

- Sign Language Translation using CNN – presented at **IEEE** conference
- Founder: Coding Club NHCE

Technical Skills

ML/DL: NumPy, ONNX, PyTorch, YOLO, CNNs, Transformers, NLP, HuggingFace, OpenCV, CUDA, Quantization, Scikit-learn
Infrastructure: Triton Inference Server, TensorRT, MetaFlow, MLflow, Docker, Kubernetes, Grafana, CI/CD
Languages & Tools: Python, C/C++, SQL, FastAPI, gRPC, Redis, FAISS, Linux, Git